

MPS-07011

Control on Process Parameters

and

Countermeasures against Process Deviation

(for TA UNIT)

Preliminary information
reference only

要因要素 原因、過程 通脱
Control on process parameters and Countermeasures against Process Deviation for TA UNIT
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Control on process parameters and Countermeasures against Process Deviation
(for TA UNIT)

Equipment	Factor	Operational Control Figure			Reason for setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
1.P-Xylene Drum (TD-101)	1) <u>Liquid Level</u>			批評			
	(1)Upper Limit (LICK-1101A)	75%	80% (Alarm)	90%	(1)P-Xylene is an organic substance which is flammable and easily frozen. For this reason, the level indicators are installed doubly in order to prevent the PX from overflowing the drum.	Level Controller with Alarm (LICK-1101A)	(1)① Compare the indication on LICK-1101A with that of LG in the field. ② Decrease an amount of P-Xylene being received from the OSBL by controlling LICK-1101A on manual. ③ Close the valve installed on the PX receiving line.
	(LI-1101B)		80% (Alarm)				
	(2)Lower Limit (LICK-1101A)	75%	70% (Alarm)	50%	(2)If P-X feed pump gets into dry-run due to a fall of level, oxygen content in the reactor rises abruptly, with a result that the operating condition of reactor would get into an explosive zone. For the said reason, it is specified to prevent the pump from getting into the dry run. (Zero(0)% indicated on the level controller is equal to the position of the suction nozzle for the P-Xylene pump.)	Level indicator with Alarm (LI-1101B)	(2)① Compare the indication on LICK-1101A with that of LG in the field. ② Increase an amount of P-Xylene being received from OSBL by controlling LICK-1101A on manual. ③ Notify Tank Yard of the trouble and request them to check the pump whether there is any trouble or not. ④ Switch the filter(TM-101) installed on the P-Xylene receiving line over to the by-pass side. ⑤ Check P-Xylene receiving line whether there is any place frozen or leaked. ⑥ Go ahead to put the unit into the ESD in case where the level of TD-101 can not be resumed to the normal figure.
	(LI-1101B)		70% (Alarm)				

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(TD-101)	2) <u>Temperature</u> Lower Limit (1) TJI-1104 Installed on inlet of TD-101	30°C	20°C	20°C	Melting point of PX is 13.3°C	Temperature Indicator with Alarm (TJI-1104)	① Move to SD operation, if P-Xylene could not be received any more.
	3) <u>Flow</u> Lower Limit (FICQ-1101)	28.9t/hr	27.9t/H (Alarm)	20t/hr	A fall of an amount of P-Xylene fed to the reactor may lead to getting into the condition with high oxygen content due to an excess supply of air. (Critical figure of 20t/H is decided on the basis of operational lower limit for the dehydrator installed on the reactor.)	Flow Controller with Alarm (FICQ-1101)	① Control FICQ-1101 on manual to adjust P-Xylene flow to be the prescribed figure. ② Check FICQ-1101 for its function, and if any problem, continue operation bypassing it. ③ Check pump TP-101 for its pumpability, and if any abnormality, switch it over to the spare side. ④ Decrease the amount of air supplied to the reactor to match an amount of P-Xylene fed to the reactor. ⑤ Move to the SD operation if normal flow of P-Xylene can not be secured in every effort.

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2.Solvent Drum (TD-102A)	1) <u>Liquid Level</u> (1)Upper Limit (LIK-1102A)	75%	85% (Alarm)	90%	Fluctuation of liquid level in the drum means the change of catalyst concentration, and leads to a retardation or promotion of reaction. For the said above, the fluctuation of liquid level should be minimized. At the upper limit of 90%, a level of TD-102A approaches to the nozzle position to which the recycle solvent is returned from the equipment concerned. On the contrary, there is a fear of causing a cavitation below the lower limit (at the level of 40% or lower).	Indicator with Alarm (LIK-1102A)	(1), (2) ① Compare the indication indicated on LIK-1102A with that of LG-1102A in the field. ② Check the amount of solvent fed to the reactor and adjust it to be appropriate figure. ③ Check the levels for TD-301 and TD-302, and correct them. ④ Check the amount of solvent for use in reslurrying in TM-300 and rinsing in TM-301, and correct them.
	(2)Lower Limit (LIK-1102A)	75%	65% (Alarm)	40%			
	2) <u>Flow Rate</u> (1)Lower Limit (FICK-1102)	143t/hr	133t/hr (Alarm)	80t/hr	A decrease in the reaction solvent leads to a retardation of oxidation reaction, resulting in a rise in oxygen content in the TA reactor.	Flow Controller with Alarm (FICK-1102)	① Adjust RQ flow rate to be normal figure by controlling FICK-1102 on manual. ② Check FICKV-1102 for its function. And, if any problem, continue operation by-passing the said valve. ③ Check the pump TP-102A or B is pumping properly. And, switch it over to the stand-by side, if any abnormality. ④ Decrease an amount of process air to be fed to the TA reactor, if necessary. ⑤ Move to the SD operation if the RQ feed to the TA reactor became impossible to be secured.

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3.Si-Oil Drum (TD-105)	1) <u>Flow Rate</u> Lower Limit (FIK-1107)	1.4kg/hr (SV)	SV-0.3 kg/hr (Alarm)	0.9kg/hr	Si-Oil is used as an anti-foaming agent for the TA reactor. For the said above, a decrease in an amount of Si-Oil fed to reactor may lead to an occurrence of a flooding phenomenon in the TA reactor.	Indicator with Alarm (FIK-1107)	① Adjust the flow rate of Si-Oil fed to the suction of TP-101 by resetting the stroke of TP-105. ② Check the CHEMICAL PUMP TP-105 for its pumpability and switch it over to the stand-by side, if any problem. ③ Check the strainer installed on the suction of FIK-1107. ④ Inspect the indication on the differential pressure guage (PDI-1204) installed on the distillation section of TA reactor, and move to SD operation in case where the stable operation could not be attained due to flooding phenomenon in the TA reactor.
4.PX Filter (TM-101)	1) <u>Differential Pressure</u> Upper Limit (PDI-1101)	1.0 kg/cm ² G $\geq$	1.3 kg/cm ² G (Alarm)	1.5 kg/cm ² G	The clogging of PX FILTER TM-101 leads to having difficulty in keeping the level of TD-101 constant. Also, the said above would damage the filter element installed in the filter in the worst case.	Differential Pressure Indicator with Alarm (PDI-1101)	① Switch the PX filter over to the by-pass side, and clean it.

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5.Purge Q Heater (TE-102)	1)Flow Rate Lower Limit (FIC-1103)	1.5t/hr	0.8t/hr (Alarm)	0.7t/hr	Agitator bearing provided for TA reactor, the stand-line with the level controller and level control valve(LICV-1201) are purged with HFQ to prevent them from being plugged with TA solid. Therefore, the decrease in an amount of HFQ above leads to a malfunction of control valve and wrong indication etc., thereby resulting in an unstable operation of TA reactor.	Flow Controller with Alarm (FIC-1103) (FIC-1104) (FIC-1105) Flow Indicator with Alarm (FI-1700)	① Adjust the flow rate of HFQ fed to user by controlling each flow controller on manual. ② Check control valve concerned for its function. And if any problem, adjust HFQ flow by switching it over to the by-pass operation. ③ Check the PURGE Q PUMP TP-103 for its pumpability. And if any abnormality, switch it over to the stand-by side. ④ Check the liquid level indicated on LIC-1208. ⑤ Move to SD operation, in case where the operation is considered to be impossible due to any leakage of pipe line and plugging etc.
	(FIC-1104)	0.5t/hr	0.4t/hr (Alarm)	0.3t/hr			
	(FIC-1105)	0.5t/hr	0.4t/hr (Alarm)	0.3t/hr			
	(FI-1700)	0.5t/hr	0.35t/hr (Alarm)	0.25t/hr			
6.TA Reactor (TD-201)	1)O ₂ Content in WG Upper limit (AISK-1201) (AISK-1202) (AISK-1203)	3.0~4.0 vol%	5.0vol% (Alarm)	8.0vol% Interlock	Due to a retardation or stoppage of oxidation reaction, oxygen content in WG rises steeply with a result that there may be a fear to get into an explosive condition. However, the gas phase comprising acetic acid, nitrogen and oxygen, even if acetic acid may be contained in any range in it, never gets into an explosive condition under an oxygen content of 10% or less. Therefore, air feed to TA reactor is to be cut at the oxygen content of 8% to avoid explosion risk.	Oxygen Analyzer with Alarm and Interlock (AISK-1201) (AISK-1202) (AISK-1203)	(1) ① Compare indications among AISK-1201, AISK-1202 and AISK-1203, and inspect analyzer concerned, if any abnormality. ② Decrease an amount of air to be fed to the TA reactor. ③ Investigate into the reactor conditions such as the level, PX flow, RQ flow, an amount of catalyst and Si-Oil flow, etc. ④ Check TA reactor for its operating parameters such as pressure (PICS-1202) and temperature (TIK-1201A), etc. (2) If two among three indicators reach the critical figure, air feed to TA reactor is stopped according to the actuation of the Interlock with its pressure and liquid maintained.

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		Normal	Allowable	Critical			
(TD-201)	2) <u>Level</u>						
	① Upper Limit						
	LIC-1201B	44%	47%	49%	(1) There is a foam layer formed above the liquid level in the reactor, and so, a rise in liquid level would lead to an occurrence of flooding phenomenon in distillation column provided on the TA reactor.	Level Controller with Alarm	(1) ① Decrease a liquid level by controlling level controller on manual. ② Compare the indication on LIC-1201B with those indicated on LICK-1201A and LG-1201. ③ Confirm that LV-1201 is functioning properly. ④ Check flow rate of Si-Oil to TA reactor to be sure that it is an appropriate one as specified by Operating Standard. ⑤ When there is no prospect of improving the disorder of discharge from the reactor, proceed to decreasing feeding PX to the TA reactor.
	LICK-1201A	44%	47% (Alarm)	49%		LIC-1201B Level Controller with Alarm	
						LICK-1201A	
	② Lower Limit				(2) It is specified to avoid a deterioration of oxidation reaction, or a rise in oxygen content in WG due to a fall of residence time.		(2) ① Raise liquid level by controlling level controller on manual. ② Compare the indication on LIC-1201B with those indicated on LICK-1201A and LG-1201. ③ Confirm that LV-1201 is functioning properly. ④ Decrease an amount of air fed to TA reactor in case where oxygen content in WG increases. ⑤ Proceed to ESD upon steep rise of oxygen content in WG.
	LIC-1201B	44%	41% (Alarm)	39%			
	LICK-1201A	44%	41% (Alarm)	39%			

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(TD-201)	3) <u>Pressure</u> (1) Lower Limit (PICK-1202)	10.5 kg/cm ² G	a) 9.5 kg/cm ² G (Alarm)	b) 7.0 kg/cm ² G Interlock	It is specified to avoid sudden flush of acetic acid due to a steep decrease in the reactor pressure. *Note Normal reactor pressure is set at appropriate one so as to attain a reactor temperature ranging 186°C to 188°C depending on the product quality.	Pressure Controller with Alarm (PICK-1202)	(1)-a) ① Adjust the reactor pressure by controlling pressure controller on manual. ② Confirm that PV-1202A/B are functioning properly. ③ Investigate into an amount of WG fed to each WG user. (GET, Pneumatic transportation etc.) (1)-b) ① SDV-1202 is closed by the actuation of the Interlock when the pressure on PICK-1202 falls down to 7.0 kg/cm ² G.
	(2) Upper Limit (PICK-1202)	10.5 kg/cm ² G	a) 10.8 kg/cm ² G (Alarm)	b) 15.0 kg/cm ² G (Design)	It is set to prevent the equipment associated with the reaction system, which is designed with 15 kg/cm ² G, from being damaged or to prevent the reactor from getting into abnormal promotion of oxidation reaction due to an increase in the reactor pressure.		(2)-a) ① Decrease the reactor pressure by controlling PICK-1202. ② Confirm that PV-1202A/B are functioning properly. (2)-b) ① Actuate the Reactor SD Interlock switch to put the reactor system into SD condition with its pressure and liquid maintained. (Note) TA reactor gets automatically into SD by the Interlock at the air flow rate of 15,000Nm ³ /hr when an air flow begins to decrease due to an increase in the reactor pressure.

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(TD-201)	4) <u>Temperature</u> (1) Upper Limit (TIK-1201A)	186~188°C	189°C (Alarm)	210°C (Design)	(1) The design of TA reactor is based on the temperature of 210°C. (2) The temperature of 170°C is the minimum one to dislocate the operating condition of reactor from the lower explosion limit.	Temperature indicator with Alarm (TIK-1201A)	(1) & (2) ① Compare the indication on LIK-1201A with those indicated on other instruments. (TJI-1211~1213, TJIK-1214~1217) ② Take the same actions as mentioned in the item of pressure other than the item ① above.
	(2) Lower Limit	186~188°C	185°C (Alarm)	170°C			
	5) <u>Air Flow</u> Lower Limit (FICSQK-1201)	114kNm ³ /hr	(1) 76kNm ³ /hr (Alarm)	20 (2) 15 kNm ³ /hr Interlock	The agitation of content in the reactor is mainly carried out by the air fed through the four lines to the bottom of reactor so that a decrease in an amount of air would lead to the deterioration of oxidation reaction, sedimentation of TA solid, plugging of discharge line and reverse flow of slurry to the process air line in the worst case.	Flow Controller with Alarm and Interlock (FICSQK-1201)	(1)-① Increase the air flow to the reactor by adjusting the flow controller on manual. ② Confirm that FCV-1209A, B, C & D are functioning properly. ③ Check TA reactor and compressor for their operating pressures. (2)-① Compressor train gets into SD by the actuation of the Interlock with a result that the reactor also enters into SD with its pressure and liquid retained.
	6) <u>Differential Pressure</u> Upper Limit (PDI-1204)	1,140 ~ 1,200 mmH ₂ O	1,250 mmH ₂ O (Alarm)	1,300 mmH ₂ O	The operating upper limit of the differential pressure indicated on PDI-1204 is specified to prevent the operation of dehydrator installed on the top of reactor from getting into a flooding phenomenon.	Differential Pressure Indicator with Alarm (PDI-1204)	① Check the level of reactor to be sure that it is operated at the normal level. ② Check an amount of Si-Oil fed to the suction of TP-101. ③ Check the reactor system to be sure that its pressure and temperature are in the range of figures as specified in the operating standard. ④ Check the amount of reflux indicated on FIK-1220 and the level indicated on LICK-1211.

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(TD-201)							⑤ Decrease an amount of PX fed to TA reactor. ⑥ Proceed to TA reactor ESD by actuating the Reactor SD switch in case where an amount of reflux on FIK-1220 and the level on LICK-1211 increase steeply. (Refer to the item of TT-201)
7.Reactor Agitator (TJ-201)	<u>Ampere</u> Upper Limit (II-1201)	17.5 ~ 25.0A	1) 30A (Alarm)	2) 48.4A (Rated Ampere)	The upper limit on ampere is specified to prevent the burning of coil due to overcurrent. Further, if the agitator stops, TA produced in the reactor, adheres to the inner wall of the reactor, thereby resulting in being impossible to continue operation.	Ampere Indicator with Alarm (II-1201)	1)-① Check oil pump and an amount of oil filled in the reducer to see if there is any problem or not. ② Check to be sure that an appropriate amount of cooling water is flowing through the agitator bearing. ③ Check the indication on FIC-1104 to be sure that the specified HFQ purge is being charged into the seal part. 2)-① In case where the agitator is expected to be not restarted soon, proceed to the ESD by actuating the Reactor SD switch.
	<u>Seal Pressure</u> Lower Limit (PICS-1207)	17kg/cm ² G	16kg/cm ² G (Alarm)	14kg/cm ² G Interlock	The mechanical seal of TJ-201 is pressurized to pressure 5kg/cm ² G higher than the operating pressure of TD-201. From view part of the fabrication for the mechanical seal, a reverse pressure on it may damage the seal part.	Pressure Indicator and Regulator with Alarm and Interlock (PICS-1207)	① Upon receiving the alarm sound, check the operating condition of PP-504 (PTA unit) and confirm that PIC-2504 and PICS-1207 are functioning properly. ② At the pressure of 14kg/cm ² G on PICS-1207, SDV-1207 is closed.

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8.1st Reactor Condenser (TE-201A)	<u>Liquid level</u> (1)Upper Limit (LICK-1221)	65%	a) 70% (Alarm)	b) 87%	There is a fear of damaging the turbine blade for TB-1132 with a mist entrained by steam, if the liquid level in the #1 Reactor Condenser rises over the specified figure. (2)If the liquid level falls down to 55% or lower, an amount of steam generated in TE-201A begins decreasing so that a power supply to TB-1132 becomes difficult. Also, the said above leads to an insufficient cooling of vapor generated from the reactor.	Level Indicator and Regulator with Alarm (LICK-1221)	1)-a ① Check the flow rate of 1BW indicated on FI-1221 and also, check the working condition of LICV-1221. ② If there is any problem with the level control valve LICV-1221, switch it over to the by-pass line to adjust the said liquid level. 1)-b)① If the level on LICK-1221 reaches 87%, LICV-1221 is closed automatically by the Interlock. ② Check TZ-1131 to be sure that the drain level is below the specified level. (2)① Take the same actions as mentioned in the items (1)-① & ② above. ② Check 1BW feed pump TP-1201 for its pumpability. If there is any problem with TP-1201, switch it over to the stand-by side. ③ Check the liquid level of TD-1201 indicated on LIC-3201.
	(LAS-1225) (2)Lower Limit (LICK-1221)	65%	60% (Alarm)	Fixed (87%) Interlock 55%		Level Switch with Interlock (LAS-1225) Level Indicator with Alarm (LICK-1221)	
9. 2nd Reactor Condenser (TE-201B)	<u>Liquid Level</u> (1)Upper Limit (LICK-1222) (LAS-1226)	70%	a) 75% (Alarm)	b) 87%	(1)If the liquid level in the #2 Reactor Condensor rises over the specified figure, a mist entrained by the generated steam increases, thereby there is a fear of damage the turbine vane for TB-1132.	Level Switch with Interlock (LAS-1226) Level Indicator with Alarm (LICK-1222)	(1)-a)① Check the working condition of LICK- 1222 and 1BW flow rate indicated on FI-1222. ② If there is any problem with LICV-1222, adjust the liquid level by-passing it. (1)-b) ① LICV-1222 is closed due to the actuation of the Interlock. ② Check TZ-1132 to be sure that the drain level is below the specified level.

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(TE-201B)	(2) Lower Limit (LICK-1222)	70%	67% (Alarm)	65%	(2) If the liquid level falls down to 65%, an amount of steam generated in TE-201B begins to decrease due to an insufficient heating area. Also, the said above leads to an insufficient cooling of vapor generated from the reactor.		(2)-① Take the same actions as mentioned in the items (1)-a) ① & ② above. ② Check liquid levels indicated on LIS-3134 and LIC-3135 installed on TE-1134, and also check TP-1131 for its pumpability. (Note) In case where TP-1131 stopped, the stand-by pump is started automatically. ③ If there is any problem with FIC-3134 and LIC-3135, adjust the liquid level in TE-1134 by-passing FIC-3134 and LIC-3135.
10. 3rd Reactor Condenser (TE-201C)	<u>Liquid Level</u> (1) Upper Limit (LICK-1223) (LAS-1227)	47%	a) 50% (Alarm)	b) 85% Fixed Point Interlock	There may be a fear of damaging the turbine blade for TB-1132 with a mist entrained by steam, if the liquid level in the #3 Reactor Condenser rises over the specified figure. If the liquid level falls down to 40% or lower, an amount of steam generated in TE-201C begins decreasing so that a power supply to TB-1132 becomes difficult. Also, the said above leads to an insufficient cooling of vapor generated from the reactor.	Level Indicator and Regulator with Alarm (LICK-1223) Level Switch with Interlock (LAS-1227) Ditto	(1)-a) ① Check the working condition of LICK-1223 and 1BW flow rate indicated on FI-1223. ② If there is any problem with LICV-1223, adjust the liquid level by-passing it. (1)-b) ① LICV-1223 is closed due to the actuation of the Interlock. ② Check TZ-1133 to be sure that the drain level is below the specified level. (2)-① Take the same actions as mentioned in the items (1)-a) ① & ② above. ② Check liquid levels indicated on LIS-3134 and LIC-3135 installed on TE-1134, and also check TP-1131 for its pumpability. (Note) In case where TP-1131 stopped, the stand-by pump is started automatically. ③ If there is any problem with FIC-3134 and LIC-3135, adjust the liquid level in TE-1134 by-passing FIC-3134 and LIC-3135.
	(2) Lower Limit (LICK-1223)	47%	43% (Alarm)	40%			

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11.Reflux Drum (TD-202)	1) <u>Liquid Level</u> (1)Upper Limit (LICK-1202)	30%	a) 40% (Alarm)	b) 80%	(1)A rise in liquid level of TD-202 worsen the gas-liquid separation efficiency, thereby increasing a mist carried over with the gas going to the vent scrubber. Further, if the said level rises to the inlet nozzle, the gas-liquid separation becomes impossible. (Note) Just below the inlet nozzle: 86% on LICK-1202	Level Indicator and Regulator with Alarm (LICK-1202)	(1)-a)① Adjust the liquid level in TD-202 by controlling LICK-1202 on manual. ② Check the reflux flow rate on FIK-1202 and an amount of DQ blown from TD-202 (indicated on FICK-1203) ③ Confirm LV-1202 is functioning properly. ④ Check TP-201C/D for its pumpability, and if any problem, switch it over to the stand-by side.
	(2)Lower Limit (LICK-1202)	30%	20% (Alarm)	10%	A fall of liquid level in TD-202 tends to be one of causes of TP-201C/D getting into running-dry operation, which disturbs the reactor operation.	Ditto	(1)-b) ① Check the differential pressure indicated on PDI-1210 for the distillation column, and if it is judged as a flooding phenomenon, immediately proceed to the ESD by actuating the Reactor SD switch. (2)-① Take the same actions as mentioned in the items (1)-a) ① through ③ above.
	2) <u>Reflux Flow Rate</u> Lower Limit (FIK-1202)	120 ~ 150t/hr	110t/hr (Alarm)	80t/hr	A fall of reflux flow to distillation column leads to insufficient removal of exothermic heat generated from the oxidation reaction, thereby resulting probably in causing flooding phenomenon.	Flow Indicator with Alarm (FIK-1202)	① Adjust the reflux flow rate on FIK-1202 by controlling LICK-1202 on manual. ② Check the liquid level of TD-202 indicated on LIC-1202. ③ Confirm that LV-1202 is functioning properly. ④ Check TP-201C/D for its pumpability, and if any abnormality, switch it over to the stand-by side.

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(TD-202)							⑤ Check the operating conditions on the reactor such as the pressure, temperature and differential pressure installed on the dehydrator section, etc.
12.Slurry Receiver (TD-203)	1) <u>Liquid Level</u> Upper Limit (LICK-1203 A/B /C)	8%	15% (Alarm)	45%	A rise of liquid level in TD-203 increases an entrainment of solid accompanied with vapor upon being flushed from the TA reactor. AS a result, there is a fear of damaging the drum due to pressurization probably caused by plugging of the heat exchanger. (Note) Just below the internal nozzle at the inlet of TD-203; 50% on LICK-1203.	Level Indicator and Regulator with Alarm (LICK-1203A/B/C)	① Adjust the liquid level of TD-203 by controlling LICK-1203A/B or C on manual. ② Confirm that LICV-1203A/B(C) are functioning properly. ③ Check TP-203 for its pumpability, and if any problem, switch it over to the stand-by side. ④ Check the suction and delivery lines of TP-203 on whether there are any pluggings in them. ⑤ Check the working condition of LV-1201 and the liquid level of TA reactor. ⑥ Check FDQ sampled from TD-204 whether it is stained or not. ⑦ Decrease PX feed to TA reactor.
	(2) <u>Pressure</u> Upper Limit (PIS-1205)	0.03~ 0.07 kg/cm ² G	(1) 0.1 kg/cm ² G (Alarm) (2) 0.15 kg/cm ² G Interlock	(3) 1.3 kg/cm ² G	TD-203 designed by the pressure of 1.8 kg/cm ² G needs to be protected from being damaged due to internal pressure rise. (Note) Rupture disc (RD-203) is installed on the vent line to protect TD-203 from being overpressurized.	Pressure Indicator with Alarm	(1)-① Check the reactor level and the working condition to be discharged through LCV-1201 to TD-203. ② Check an amount of FDQ sprayed on the surface of tube sheet for TE-203A. ③ Check the bottom of TM-203 whether there is any plugging or not. (2)-① TZ-201 is closed by the actuation of the Interlock. However, after the indication on PIS-1205 has been returned to normal condition, TZ-201 can be opened.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TD-203)							(3)-① In case where the pressure indicated on PIS-1205 is not returned to normal figure soon, immediately proceed to ESD by actuating the Reactor SD Switch.
13.Slurry Receiver Agitator (TJ-203)	<u>Ampere</u> Upper Limit (II-1203)	10~11A	(1) 16A (Alarm)	(2) 23.8A (Rated)	The upper limit on ampere is specified to prevent the burning of coil due to overcurrent. Further, if the agitator stops, TA discharged from the TA reactor precipitates immediately and accumulates in TD-203, thereby resulting in being impossible to continue operation.	Ampere Indicator with Alarm (II-1203)	(1)-① Check the liquid level of TD-203 ② Check oil level filled in the reducer of TJ-203. ③ Check an amount of FQ charged into the seal part of lower bearing. (2)-① In case where TJ-203 can not be restarted soon, proceed to ESD by actuating the Reactor SD switch.
14.Flash Condenser (TE-203 A/B)	<u>Temperature</u> Upper Limit (1)TJI-1207 (TD-204) (2)TJI-3345 (RCCW)	60~70°C 39~42°C	70°C (Alarm) 49°C (Alarm)	110°C 60°C (Design)	(1)The upper limit on the temperature for the process side is decided on the basis of the temperature at the outlet of TE-203B, which is equivalent to the dew point of vapor in TE-203B. (2)The upper limit on the temperature for the RCCW exiting from TE-203B is set at 60°C based on the shell design temperature of TE-203B.	Temperature Indicator with Alarm (TJI-1207) Temperature Indicator with Alarm (TJI-3345)	(1) & (2) ① Check the feed rate of 1BW fed to TE-203A based on the indications on FIC-3134, FI-1222 and FI-1223. ② Check the flow rate of CCW fed to TE-203B and its temperature. ③ Check the variation of an amount of slurry discharged through LCV-1201 to TD-203. ④ In case where 1BW fed to TE-203A was stopped, immediately increase an amount of CCW fed to TE-203B.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
15. Waste Gas Scrubber (TT-202)	1) <u>Liquid Level</u>				A rise of liquid level in TT-202 worsen the gas-liquid separation efficiency, thereby increasing a mist entrained with WG. (Note) Just below the inlet nozzle; 90% on LIC-1206A	Level Indicator and Regulator with Alarm (LIC-1206A/B)	① Adjust the liquid level by controlling LIC-1206A/B on manual. ② Check LCV-1206A/B for its function ③ Check an amount of liquid discharged through FIK-1204. ④ Check the temperature of TD-202 on TI-1202 and the liquid level on LICK-1202.
	Upper Limit (LIC-1206A)	50%	65% (Alarm)	90%			
	(LIC-1206B)	50%	65% (Alarm)	90%			
	2) <u>Amount of DIW</u>				A decrease of DIW fed to TT-202 increases an amount of acetic acid contained in WG discharged from TT-202.	Flow indication with alarm (FIC-1216)	① Check TP-202C(D) for its pumpability, and if any abnormality, switch it over to the stand-by pump. ② Check the pressure of DIW indicated on PI-3300.
	Upper Limit (FIC-1216)	1900kg/hr	2000kg/hr	2100kg/hr			
	Lower Limit	1900kg/hr	1800kg/hr	1700kg/hr			
	3) <u>Amount of FDQ</u>				A decrease of FDQ fed to TT-202 increases an amount of TBE (Tetra Bromethene) contained in WG discharged from TT-202.	Flow indication with alarm (FIC-1213)	① Check TP-202A(B) for its pumpability, and if any abnormality, switch it over to the stand-by pump. ② Check TP-204 for its pumpability, and if any abnormality, switch it over to the stand-by pump.
	Upper Limit (FIC-1213)	9000kg/hr	10000kg/hr	11000kg/hr			
	Lower Limit	9000kg/hr	8000kg/hr	7000kg/hr			

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
16.Waste Gas Cooler (TE-202)	<u>Temperature</u> Upper Limit (1)TJI-1208 (TE-202BTM)	35~40°C	45°C (Alarm)	60°C (Design)	(1)The upper limit on the temperature for the bottom of TE-202 is set at the design temperature of TT-202.	Temperature Indicator with Alarm (TJI-1208)	(1) ① Check the flow rate and the temperature of CCW fed to TE-202. ② Check the variation of load to TE-202 based on such parameters as the temperature and liquid level of TD-202, and the bottoms of TT-202 indicated on FI-1204 and FI-1214, etc. ③ Increase an amount of CCW fed to TE-202
17. Centrifuge (TM-300 A/B/C)	1) <u>Torque</u> Upper Limit (XJI-1300 A/B/C)	50~75 kg·m	(1) 100 kg·m (Alarm)	(2) 170 kg·m Interlock	The upper limit on the torque is specified to prevent the conveyor and gear box of the centrifuge from being damaged due to overload.	Torque Indicator with Alarm and Interlock (XJI-1300A/B/C) Torque Limitter with Interlock (ZS-1300A/B/C)	(1)-① Decrease an amount of slurry fed to the centrifuge to confirm a decrease in the torque indicated on XJI-1300. ② Switch centrifuge over to the stand-by side and wash it with caustic, in case where the torque value doesn't decrease even with the action above. (2)-① When the torque value reaches 170 kg·m or larger, the torque arm trips the machine. As a result, slurry fed to the centrifuge is cut according to the Interlock. ② Start the stand-by centrifuge and repair the machine tripped.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TM-300 A/B/C)	2) <u>Lube Oil</u>						
	(1) Lower Limit for LO Pressure	kg/cm ² G	a) kg/cm ² G (Alarm)	b) kg/cm ² G Interlock	(1) They are specified to prevent the centrifuge from being damaged due to a shortage of lube oil.	• Pressure Switch with Interlock (PS-3522A/B/C)	(1)-a) ① Check the flow rate of LO fed to the Spindle, Gear Box and Seal Coolant, and their LO temperatures.
	• Spindle Oil (PS-3522 A/B/C)	0.7~1.5		0.5		• Pressure Indicator with Alarm and Interlock (PIS-3523A/B/C)	② Check LO line whether there is any leakage or not.
	• Gear Box Oil (PIS-3523 A/B/C)	3.0~5.0	2.0	1.0		• Pressure Indicator with Alarm and Interlock (PIS-3524A/B/C)	③ Confirm that LO pump is functioning properly.
	• Seal Coolant Oil (PIS-3524 A/B/C)	1.0~2.0	1.0	0.5			④ Check the Oil levels indicated on LI-3521 A/B/C.
	(2) <u>LO Leakage</u>						
	Upper Limit (LI-3521A/B/C)	5~8 ml/hr ≥	5.8ml/hr	4l/8hrs	It is specified to prevent the damage of machine as well as the degradation of product due to a leakage of lube oil.	• Oil level in oil tank is to be checked every 4 hours. • Level Indicator with Alarm (LI-3521A/B/C)	(1)-b) ① Main motor is stopped and also slurry feed is shut by the Interlock. ② Start the stand-by centrifuge. (2)-① In case where the oil leakage is 4l/8hrs or more, switch TM-300 over to the stand-by side, and repair it.
	(3) <u>Oil Level</u>						
	Lower Limit (LI-3521A/B/C)	60~70%	50% (Alarm)	40%	Ditto	Ditto	(3)-① Replenish LO tank with oil.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TM-300 A/B/C)	3) <u>Vibration</u> Upper Limit (VJI-1300 A/B/C)	50~150 μ	(1) 180 μ (Alarm)	(2) 200 μ	It is specified to prevent the machine from damaging based on the vibration value of 200 μ guaranteed by the vendor.	Vibration Indicator with Alarm (VJI-1300A/B/C)	(1)-① Adjust an amount of slurry fed to the centrifuge ② Check plugging of lower chute and ML line of centrifuge with TA ③ Wash the centrifuge with caustic (2)-① When the vibration reaches 200 μ or larger, switch the centrifuge over to the stand-by side and repair it.
	4) <u>Main Motor</u> Ampere Upper Limit (IIK-1300 A/B/C)	27~35A	(1) 35A (Alarm)	(2) 38A (Rated)	The upper limit on ampere is specified to prevent the burning of coil due to overcurrent.	Ampere Indicator with Alarm and Interlock (IIK-1300A/B/C)	(1) Take the same actions as those mentioned in the items 3)-(1) ①, ② & ③ above. (2) In case where motor was stopped due to overload; ① The slurry feed to the centrifuge is stopped according to the Interlock. ② Start the stand-by centrifuge.
18. Reslurry Drum (TD-300)	Liquid Level (1) Upper Limit (LICK-1300A/B)	10%	50% (Alarm)	90%	Since a rise in liquid level of TD-300 leads to a plugging of lower chute of the centrifuge, thereby resulting in discontinuation of plant operation, the upper limit on the level is specified at 90%, which is also the upper limit of LG to be readable.	Level Indicator and Regulator with Alarm (LICK-1300A/B)	(1)-① Adjust the slurry feed to TM-301 by controlling LICK-1300A/B on manual. ② Check FICK-1300A/B or C to be sure that the prescribed amount of acetic acid is being supplied to centrifuge. ③ Check to be sure whether there is any ML fell from the bottom of centrifuge or not. ④ Check TP-300 to see if there is any plugging on the suction and delivery lines of it. ⑤ Wash RVF with caustic if it is in disorder.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TD-300)	(2) Lower Limit (LICK-1300A/B)	10%	2% (Alarm)	2%	(2) It is specified to prevent the pump from getting into the cavitation condition.		(2)-① Adjust the slurry feed to TM-301 by controlling LICK-1300A/B on manual. ② Check an amount of acetic acid to be fed for reslurrying TA in TD-300 (FICK-1300A/B or C) ③ Check equipment on the upstream whether there is any plugging or not.
19. Reslurry Drum Agitator (TJ-300)	<u>Ampere</u> Upper Limit	7.5A	(1) 15A	(2) 17.9A (Rated)	The upper limit on ampere is specified to prevent the burning of coil due to overcurrent. Further, the stoppage of agitator leads to precipitation of solid in suspension and accumulation of solid in the drum.	Ampere Indicator with Alarm	(1)-① Check liquid level in the drum. ② Check an amount of oil filled in the reducer. ③ Confirm the amount of acetic acid on FICK-1300A/B or C for use in reslurrying TA separated in TM-300. (2)-① In case where the agitator stopped due to overload can not be restarted soon; · Inject N ₂ at the bottom of drum to prevent slurry from precipitating · Decrease PX feed to TA reactor or proceed to ESD in the worst case
20. ML Drum (TD-301)	1) <u>Liquid Level</u> Upper Limit (LIC-1301)	10%	65% (Alarm)	90%	Since a rise in liquid level of TD-301 leads to a trouble discharging ML separated in centrifuge, thereby resulting in discontinuation of plant operation, the upper limit on the level is specified at 90%, which is below the inlet nozzle of ML and SG, and close to the upper limit of LG to be readable.	Level Indicator and Regulator with Alarm (LIC-1301)	① Adjust the liquid level by controlling LICK-1301 on manual ② Confirm that FICV-1310 is functioning properly ③ Check TP-301 for its pumpability, and if any problem, switch it over to the stand-by side. ④ Check plugging on both suction and delivery lines of TP-301. ⑤ In case where the liquid level in TD-301 can not be reduced, decrease PX to TA reactor or proceed to the ESD.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TD-301)	(2) <u>ML Blow</u> Lower Limit (FICK-1311)	6.8t/hr	6.3t/hr (Alarm)	2.5t/hr	A decrease in an amount of ML blown to TM-501 leads to an accumulation of oxidation inhibitor in the recycle solvent, thereby deteriorating the product quality.	Flow Indicator and Regulator with Alarm (FICK-1311)	① Adjust the ML flow rate blown to TM-501 by controlling FICK-1311 on manual. ② Check the plugging in the ML-Blow line and also, check TP-301 for any abnormality. ③ Wash ML-Blow line with caustic.
21.RVF (TM-301 A/B)	1) <u>Liquid Level</u> Upper Limit (LIS-1304A/B)	70~80%	(1) 90% (Alarm)	(2) 95% Interlock	If the overflow line from TM-301 to TD-300 is plugged, liquid level rises in the feed pan so that a liquid overflows the pan to the dryer. Note; The liquid level of 90% is located at the place, 110mm lower than the top of weir.	Level Indicator with Alarm and Interlock (LIS-1304A/B)	(1)-① Check the indication on LIS-1304A/B for its correctness. ② Check slurry feed to TM-301A/B indicated on FICK-1301A/B. ③ Check plugging in the overflow line and purge it, if any abnormality. (2)-① According to the actuation of Interlock, slurry feed to TM-301A/B (FCV-1301A/B), FDQ rinse to TA cake (FCV-1302A/B) and the filter-cloth "wash sequence" are automatically stopped. ② If the plugging in the overflow line can not be solved even with the FDQ purge, wash it with caustic. (Note; Slurry is to be stored in TD-300 during the said caustic wash)

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TM-301 A/B)	2) <u>Ampere</u> Upper Limit (IJIK-1301A/B)	5-10A	12A (Alarm)	13A (Rated)	The rated ampere is specified as the critical figure to prevent the motor from damaging	Ampere Indicator with Alarm (IJIK-1301A/B)	① Check the slurry feed rate on FICK-1301A/B and revolution speed on XIK-1301A/B ② Check the motor and the body of TM-301 for any abnormalities such as vibration and abnormal sound, etc. ③ If one of RVFs became malfunction, stop it and repair.
	3) <u>Vaccum</u> Lower Limit (PICK-1302)	-150 ~ -400mmHg	-550mmHg (Alarm)	-600mmHg	It is specified to monitor the degree of plugging on the filter cloth.	Pressure Indicator and Regulator with Alarm (PICK-1302)	① Check the indication on PICK-1302 for its correctness by comparing it with that on PG-1303. ② If the capacity decreased to the specified level due to plugging of filter cloth, wash RVF with caustic.
	4) <u>Blow Back Pressure</u> (1) Upper Limit (PICK-1301A/B)	0.12 kg/cm ² G	0.17 kg/cm ² G (Alarm)	0.2 kg/cm ² G	It is specified to prevent a damage of filter cloth due to a high pressure.	Pressure Indicator and Regulator with Alarm (PICK-1301A/B)	(1)- ① Adjust the blow-back pressure by controlling PICK-1301A/B on manual, if necessary. ② Check TC-303 for its abnormality. ③ If there is any problem with PICV-1301A(B), etc., repair it .
	(2) Lower Limit (PICK-1301A/B)	0.12 kg/cm ² G	0.11 kg/cm ² G	0.09 kg/cm ² G	It is specified to prevent a TA cake on filter cloth from being left over due to a low blow-back pressure.		(2)-① Take the same action as mentioned in the item (1)-① above.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
22. Filtrate separating Drum. (TD-302)	<u>Liquid Level</u> Upper Limit (LICS-1302)	15%	(1) 25% (Alarm)	(2) 35% Interlock	When the liquid level of TD-302 rises, the filtrate discharged from TM-301 is to be obstructed with a result that it leads to a decrease in an amount of gas sucked in TC-303, and the overload of TC-303 due to liquid entrained with gas in the worst case. Also, if the filtrate enter TD-303, FDQ in TD-204 is fouled with catalyst. (Note) The position just below the filtrate inlet nozzle on TD-302 is equivalent to approx. 36% on LICS-1302.	Level Indicator and Regulator with Alarm (LICS-1302)	(1)-① Check the indication on LICS-1302 for its correctness by comparing it with that on LG-1302. ② Check the working condition of LICV-1302, and adjust the liquid level by controlling LICK-1302 on manual. ③ If there is any plugging on the suction of TP-302, switch it over to the side suction and wash it with caustic. (2)-① At 35% indicated on LICS-1302, PICV-1302 is opened fully, according to the Interlock, thereby resulting in the breach of vacuum condition. ② If the plugging on the suction of TP-302 can not be solved even with caustic wash, stop feeding slurry to TM-301 and wash it with caustic perfectly.
23. Vacuum Pump (TC-303 A/B/C)	1) <u>Ampere</u> Upper Limit (IJI-1301 A/B/C)	~175A	195A (Alarm)	198A (Rated)	The rated ampere is specified as the critical figure to prevent the damage of TC-303.	Ampere Indicator with Alarm	① Adjust the flow rate of sealant to TC-303A/B or C to be the specified ones by controlling FIC-1303A/B or C respectively ② Check TC-303 for abnormal sound, vibration, tension of belt and temperature rise of motor, etc. ③ If the said TC-303 is judged to have any problem, switch it over to the stand-by side.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TC-303 A/B/C)	2) <u>Sealant Flow</u> (1) Upper Limit (FIC-1303 A/B/C)	25t/hr (Per one unit)	28t/hr (Alarm)	30t/hr	(1) If an excessive amount of sealant is fed to TC-303, there is a case to cause such troubles as an abnormal sound and vibration, etc. resulting in overload.	Flow Indicator and Regulator with Alarm (FIC-1303A/B/C)	(1)- ① Check FCV-1303A/B or C for their function, and adjust the flow rate by controlling them on manual. ② Check the pressure indicated on PG-1204A(B) on the outlet of TP-204A(B).
	(2) Lower Limit (FIC-1303 A/B/C)	25t/hr	18t/hr (Alarm)		It is specified as the minimum flow required for removing the compression heat, which is generated by TC-303 in operation.	Ditto	(2) Take the same actions as mentioned in the item (1) above.
24. Vacuum Pump Separating Drum (TD-303)	<u>Liquid Level</u> (LIC-1303)	20%	30% (Alarm)	40%	If the liquid level of TD-303 rises, the outlet line of TC-303 is obstructed with a result that it leads to a decrease in the gas discharged from TC-303 and the overload of TC-303 in the worst case. (Note) The position just below the gas inlet nozzle on TD-303 is equivalent to the level of approx. 40% indicated on LIC-1303.	Level Indicator and Regulator with Alarm (LIC-1303)	① Check the indication on LIC-1303 for its correctness by comparing it with that on LG-1303. ② Check the working condition of LCV-1303, and adjust the liquid level by controlling LIC-1303 on manual. ③ Check the flow rate of sealant fed to TC-303A/B or C on FIC-1303A/B or C.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
25.Dryer (TM-304)	1) <u>Temperature</u> Lower Limit (TI-1310)	136°C	132°C (Alarm)	131°C	A fall of temperature on the outlet of dryer leads to an increase in the volatile content in the TA product.	Temperature Indicator with Alarm (TI-1310)	① Increase the set pressure of thermo compressor through PIC-1304. ② Confirm the steam trap (ST-1304) is functioning correctly. ③ Check RVF whether there is any liquid overflowing the weir to the dryer or not, and wash RVF with caustic, if any overflow. ④ Check circulating N ₂ gas flow indicated on FI-1306 and its temperature indicated on TJI-1315. ⑤ In case where the heating tube of dryer has been fouled; a) Stop feeding slurry to RVF, operate the dryer under no load condition, paying attention to the temperature on TI-1310. b) If the performance of dryer has not been restored to the normal condition in spite of no- load operation above, wash the dryer with caustic.
	2) <u>Steam Flow</u> Upper Limit (FI-1304)	3.0-4.0 t/hr	4.0t/hr (Alarm)	6.0t/hr	An increase in an amount of heating steam means an increase of wetness contained in wet cake and an introduction of slurry overflowing the weir to the dryer. Accordingly, it is specified to quickly detect the abnormal conditions and to prevent a deterioration of product quality.	Flow Indicator with Alarm (FI-1304)	① Check steam pressure on PIC-1304. ② Check TM-301 whether there is any liquid overflowing the weir to the dryer, and if any, wash it with caustic. ③ Check TM-301 whether there is any damage on the filter cloth or not. ④ Check the liquid level of TT-301 to see if there is any overflow to dryer. ⑤ Confirm that the indication on FI-1304 is correct, and that the steam trap is working properly.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TM-304)							⑥ Check the operating conditions for the dryer system whether there is any damage on the heating tube or not. If any damage on the heating tube, move to the ESD.
	3) <u>O₂ content</u> Upper Limit (AI-1300A)	4-5vol%	7vol% (Alarm)	8vol%	TA powder in the gas containing high concentration of O ₂ has a possibility of causing dust explosion. And so, the said oxygen content must be kept at 8% or lower.	O ₂ Content Indicator with Alarm (AI-1300A)	① Check N ₂ flow to the labyrinth of TC-301. ② Check the liquid sealed in the U-seal at the bottom of TM-305. ③ Check the dryer pressure and the working condition of PICV-1306. ④ Confirm that the analyzer (AI-1300A) and the scrubbing unit are functioning properly. ⑤ Check TC-301 to be sure that there is not air sucked in the dryer system through the suction of it.
	4) <u>Dryer Pressure</u> Upper Limit (PIC-1306)	50mmAq	200mmAq (Alarm)	800mmHg (Seal Pot)	The upper limit on the dryer pressure is specified to prevent acetic acid vapor and TA from spouting from the dryer system owing to an excess pressure. (Note) Dryer designed by the pressure of 0.1 kg/cm ² G is protected by the seal pot having the sea leg of 800mmAq in depth.	Pressure Indicator and Regulator with Alarm (PIC-1306)	① Adjust the dryer pressure by controlling PIC-1306 on manual ② Confirm that PCV-1306 is functioning correctly. ③ Check TT-203A system to be sure that there is no plugging around the system. ④ Check N ₂ flow to TC-301 labyrinth to be sure that it is flowing at the specified rate. ⑤ Confirm TM-301A/B to see if they are operated in a good condition. ⑥ Confirm that the purge sequence conducted for flushing the line between TM-302 and TT-301 is being carried out correctly, and that there is no plugging in the said line. ⑦ Check the liquid level in TT-301 to see whether there is any overflow to dryer or not.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TM-304)	5) <u>Motor Ampere</u> Upper Limit (1) TM-304-1 (IJI-1304A)	49~52A	a) 60A (Alarm)	b) 96A (Rated)	(1), (2), (3) The upper limit on the motor ampere is specified to prevent motor from being damaged due to its overload. And also, the Interlock is provided to prevent the dryer system from being plugged with TA upon its stoppage.	• Ampere Indicator with Alarm (IJI-1304A) • Stop Alarm with Interlock	(1)-a)-① Adjust the slurry feed to TM-301 to be appropriate one. ② Check RV (TZ-304) installed on the outlet of dryer to be sure that it is functioning correctly. ③ Confirm that the knockers installed on the surface of TM-302 are functioning correctly. (1)-b)-① According to the actuation of Interlock, equipments installed in the upstream of the dryer system are stopped. a) Decrease an amount of PX fed to the reactor and try to restart the dryer system. b) In case where the dryer system can not be restarted soon, proceed to the ESD of the unit.
	(2) TM-304-2 (IJI-1304B)	6~7A	a) 12A (Alarm)	b) 17.8A (Rated)			
	(3) TZ-304	3~3.2A	a) 6.4A (Rated)	b) 6.4A (Rated)			
						• Ampere Indicator with Alarm (IJI-1304B) • Stop Alarm with Interlock	(2)-a) Take the same actions as mentioned in the item 1)-a) above. (2)-b) Take the same actions as mentioned in the item 1)-b) above.
						• Stop Alarm with Interlock	(3)-a)-① Check TZ-304 for its stroke and adjust the stroke to be appropriate one. ② Confirm that the slurry feed to TM-301 is an appropriate level.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TM-304)							(3)-b)-① Due to the actuation of the Interlock upon its stoppage, all equipments in the upstream of the dryer system are stopped. a) Decrease an amount of PX fed to the TA reactor and try to restart the dryer system. b) In case where the dryer system can not be restarted soon, proceed to the ESD of the unit.
26. TA Dryer N ₂ Circulation Blower (TC-301 A/B)	1) <u>Carrier Gas Flow</u> (1) Upper Limit (FI-1306) (2) Lower Limit (FI-1306)	4,864 Nm ³ /hr 4,864 Nm ³ /hr	5,650 Nm ³ /hr (Alarm) 5,200 Nm ³ /hr (Alarm)	5,800 Nm ³ /hr (Design) 5,000 Nm ³ /hr	An increase in an amount of circulated N ₂ gas increases both drying speed and capacity. However, an amount of fine particles of TA entrained increases, thereby resulting in causing a plugging phenomenon at TT-301. A shortage of amount of carrier gas leads to an inferior drying of product.	Flow Indicator with Alarm (FI-1306) Ditto	(1)-① Adjust the flow rate on FI-1306 by controlling the TC-301 suction valve (HC-1307) ② Check the dryer internal pressure to be sure that PIC-3106 is functioning correctly. ③ Check the slurry feed rate to TM-301A/B indicated on FIK-1301A/B. (2)-① Take the same actions as mentioned in the item (1) above. ② Switch TC-301 over to the stand-by side. ③ Check to be sure that the purge to the line connecting TM-302 with TT-301 is being carried out correctly according to the sequence, and that the said line is not plugged with TA crystallized. ④ Confirm that the seal leg in TZ-306 is sealed completely with water. ⑤ Check to be sure that there is no plugging in TT-301 trays and TE-301 tubes.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TC-301 A/B)	2) <u>Ampere</u> Upper Limit (Indicator on the Field)	~50A	(1) 58A (Rated)	(2) 58A (Rated)	The upper limit on the motor ampere is specified to prevent motor coil from damaging due to its overcurrent.	Stop Alarm with Interlock	(1)-① Adjust the N ₂ gas flow to the dryer by controlling HC-1307. ② Switch TC-301 over to the stand-by side. (2)-① If TC-301 stops due to overload, all equipments in the upstream of it for the dryer system are to be stopped according to the Interlock. a) Start the TC-301 on stand-by and restart the dryer system in accordance with the manual.
27. Dryer N ₂ Scrubber (TT-301)	1) <u>Liquid Level</u> Upper Limit (LICS-1305)	50%	a) 60% (Alarm)	b) 80% Interlock	A rise in liquid level of TT-301 leads to an overflow of liquid through TM-302 to the dryer. (Note) The level of 92% on LICS-1305 is equivalent to that just below the nozzle on TT-301, which connects TT-301 with TM-302.	Level Indicator with Alarm (LICS-1305)	a)-① Adjust the liquid level in TT-301 by controlling LICS-1305 on manual. ② Check TP-310 to see if there is any plugging in both suction and delivery lines of it. ③ If any problem with TP-310, switch the line to TD-701 and repair it. ④ Check an amount of FDQ fed to TT-301 through FIC-1307. b)-① When the level on LICS-1305 reaches 80%, FIC-1307 is closed according to the actuation of the Interlock. (Note; At the level of 50% on LICS-1305, those mentioned above are automatically returned to the normal conditions)

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TT-301)	2) <u>FDQ to TT-301</u> Lower Limit (FIC-1307)	10t/hr	5t/hr (Alarm)	3t/hr	A decrease in the amount of FDQ fed to TT-301 means a fall of collection of acetic acid and fine particle entrained. Also, the said above becomes one of the causes of plugging trays in TT-301 with solid deposited.		① Adjust the FDQ flow to TT-301 by controlling FIC-1307 on manual. ② Confirm that FICV-1307 is functioning correctly. ③ Check the pump TP-204 for its pumpability. Switch it over to the stand-by side, if any problem.
28.Silo (TTK-400) (TTK-401)	1) <u>WG Flow</u> Lower Limit (1) FIC-1305 (TM-304 → TTK-400) (2) FIC-1401 (TTK-400 → PTK-101) (3) FIC-1402 (TZ-413 → TTK-401) (TTK-401 → TTK-400)	Nm ³ /hr 2,200 2,200 400	a) Nm ³ /hr (Alarm) 2,000 2,000 350	b) Nm ³ /hr Interlock 1,800 1,800 300	The lower limit on the waste gas flow is specified to prevent the plugging in the pneumatic transportation line due to a shortage of the WG flow.	Flow Indicator and Regulator with Alarm and Interlock (FIC-1305) (FIC-1401) (FIC-1402)	(1), (2), (3) a)- ① Adjust the WG flow by controlling each flow controller on manual. ② Confirm that the flow control valve concerned is functioning Properly. ③ Check the dryer system to be sure that there is no variation of amount of TA transported with pneumatic transportation. ④ Check to be sure whether the pneumatic transportation line is plugged with TA or not. ⑤ Check that the bag filter is not plugged with TA solid. ⑥ Confirm the pressure of waste gas for use in pneumatic transportation (PIC-3104) b)- ① According to the actuation of the Interlock, TA product fed to the pneumatic transportation line is stopped. ② In case where the waste gas can not be used any more, switch it over to the pneumatic transportation with the blower being in operation.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TTK-400) (TTK-401)	2) Pressure						
	(1) Upper Limit a) PIC-1400 (TM-403 Inlet)	70-90 mmAq	300mmAq (Alarm)	800mmAq (PRV)	The upper limit on the silo pressure is specified to prevent the silos from being damaged due to overpressure because the silos are designed by the pressure of 0.1 kg/cm²G. Note; Critical figure is based on the relieving pressure of PRV(800mmAq) • PRV-1400 ... TTK-400 • PRV-1401 ... TTK-401	Pressure Indicator and Regulator with Alarm PIC-1400	(1)- ① Adjust the pressure of silo by controlling PIC-1400 on manual. ② Confirm that the pressure control valve PICV-1400 is functioning correctly. ③ Confirm that an appropriate amount of N₂ is introduced through the aerator installed at the bottom of silo. ④ Check that the bag filter is not plugged with TA solid. ⑤ Switch silo from TTK-400 to TTK-401 and vice versa.
	(2) Lower Limit (PICS-1404)	200mmAq	a) -30mmAq (Alarm)	b) -40mmAq Interlock	While the blower is being operated, there is a possibility that the silo is damaged due to a vacuum condition probably resulted from the plugging of the bag filter, etc. (Note) The silo is designed at -100mmAq on the vacuum side.	Pressure Indicator and Regulator with Alarm and Interlock (PICS-1404)	(2)-a)- ① Check the differential pressure on PDI-1403 for the bag filter. ② Check pneumatic flow rate and reset it, if necessary. ③ Check pressure control valves to be sure that they are functioning correctly. b)- ① According to the actuation of the Interlock, blower(TC-302) is stopped.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TTK-400) (TTK-401)	3) <u>O₂ Concentration</u> Upper Limit (AI-1300B)	4vol%	7vol% (Alarm)	8vol%	TA powder in the gas containing high concentration of O ₂ has a possibility of causing dust explosion. And so, the said oxygen content must be kept at 8% or lower.	O ₂ Content Indicator with Alarm (AI-1300B)	① Check that the O ₂ analyzer (AI-1300B) and the scrubbing unit are functioning correctly. ② In case where the blower is operated; a) Check PIC-1400 to be sure that the pressure for silo system is being controlled in a good condition. b) Adjust the pneumatic gas flow by controlling the by-pass valve of TC-302. c) Check the bag filter to be sure that the differential pressure indicated on PDI-1403 is within the allowable range. d) Check blower to be sure that there is no air sucked from the suction of blower.
29. Bag Filter (TM-403)	<u>Differential Pressure</u> Upper Limit (PDI-1403)	40~70 mmAq	130mmAq (Alarm)	250mmAq	The filter cloth and filter retainer may be collapsed by the differential pressure of 250mmAq or higher.	Differential Pressure Indicator with Alarm (PDI-1403)	① Check to be sure that the waste gas fed to the pneumatic transportation line is in an appropriate range. ② Check the control unit in the field to be sure that the N ₂ pressure on the pulse unit is an appropriate and that the pulse unit is functioning properly. ③ Replace the filter cloth with the new one, if it is considered the end of life.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
30.FQ Drum (TD-208)	<u>Liquid Level</u> Lower Limit (LIC-1208)	46%	35% (Alarm)	20%	Pump run dry due to a low level in TD-208 can not supply purge to the mechanical seal and instrument, thereby resulting in causing a malfunction of instrument and damage of equipment, etc.	Level Indicator and Regulator with Alarm (LIC-1208)	① Adjust the liquid level in TD-208 by controlling LIC-1208 on manual. ② Confirm that the level controllers LIC-1208A/B are functioning properly. ③ Check an amount of acetic acid entering TD-208 as mentioned below; a)TD-204 bottoms indicated on LIC-1204. b)Make-up indicated on FIC-1218, which is sent from TK-421. c)Acetic acid condensed in TE-504 and TE-505, which is confirmed through FIK-1554 and FIK-1555. ④ Make-up indicated on FIC-1215, which is sent from TD-201, also check to TD-208 from TD-207. ⑤ Check an amount of acetic acid discharged from TD-208. a)An amount of HFQ indicated on FIC-1103, FIC-1104, FIC-1105 and FI-1700. b)An amount of FQ indicated on FIK-1701 and FI-1205.
31.First Film Evaporator (TM-501)	1) <u>Ampere</u> Upper Limit (IJI-1501)	35~50A	(1) 50A (Alarm)	(2) 115A (Rated)	The upper limit on the motor ampere is specified to prevent motor coil from damaging due to its overcurrent. In addition, TM-501 is provided with the Interlock to actuate upon its stoppage from view point of preventing it from giving influence on the process side.	Ampere Indicator with Alarm (IJI-1501) Stop Alarm with Interlock (SD-1501)	(1)-① Check the feed rate on FICK-1311, and adjust it, if necessary. ② Check to see if the bottom line of TM-501 is through, and purge it with caustic or steam, if any plugging. ③ Check TM-501 whether there are such abnormalities as abnormal sound, vibration and high temperature, etc. ④ Wash TM-501 with caustic.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TM-501)							(2)-① In case where TM-501 stopped, the following valves are closed by the actuation of the Interlock. a)ML feed to TM-501. (FCV-1311) b)Heating steam to the jacket of TM-501. (FCV-1551) c)Heating steam to the jacket of TM-502. (FCV-1552)
32.Second Film Evaporator (TM-502)	1) <u>Ampere</u> Upper Limit (IJI-1502)	10~12A	(1) 20A (Alarm)	(2) 31A (Rated)	The vapor limit on the motor ampere is specified to prevent motor coil from damaging due to its overcurrent. In addition, TM-502 is provided with the Interlock to actuate upon its stoppage from view point of preventing it from giving influence on the process side.	Ampere Indicator with Alarm (IJI-1502) Stop Alarm with Interlock (SD-1501)	(1)-① Check ML feed rate on FICK-1311. ② Check to see if the bottom line of TM-502 is through, and purge it with caustic or steam, if any plugging. ③ Check TM-502 whether there are such abnormalities as abnormal sound, vibration and high temperature, etc. ④ Wash TM-502 with caustic. (2)-① In case where TM-502 stopped, the same action as mentioned in the item (2)-① above are carried out by the actuation of the Interlock.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
33. Centrifuge (TM-503)	1) <u>Torque</u> Upper Limit (XI-1503)	1-7kg · m	(1) 10kg · m (Alarm)	(2) 14kg · m Interlock	The upper limit on torque is specified to prevent the conveyor and gear box of the centrifuge from being damaged due to overload.	Torque indicator with Alarm and Interlock (XI-1503) Toque Limitter with Interlock (ZS-1503)	(1)-① Decrease an amount of slurry fed to TM-503 by controlling LICK-1505 on manual ② In case where the torque value can not be reduced even with the action said above, wash TM-503 with caustic. (2)-① When the torque value reaches 14kg · m or larger, the main motor is tripped, and slurry feed valve (LCV-1505) and feed tube valve (KCV-1584) are closed by the actuation of the Interlock.
	2) <u>LO Pressure</u> Lower Limit (1) Gear Box (PS-3541) (2) Spindle Oil (PS-3543) (3) Seal Coolant (PS-3544)	kg/cm ² G 2.7 2.7 4.7		kg/cm ² G 1.0 Interlock 0.5 Interlock 0.5 Interlock	They are specified to prevent centrifuge from being damaged due to the decrease in the LO pressure.	Pressure Switch with Interlock (PS-3541) (PS-3543) (PS-3544)	(1), (2), (3) By the actuation of the Interlock, the main motor is tripped, and slurry feed valve (LCV-1505) and feed tube valve (KCV-1584) are closed automatically.
34. 3% NaOH Drum (TD-1421)	<u>Liquid Level</u> Upper Limit (LIK-3421)	20-80%	85% (Alarm)	90%	It is specified to prevent NaOH causing dermatitis from overflowing due to an excessive rise of liquid level.	Level Indicator with Alarm (LISK-3421)	① Confirm that 3% NaOH Preparation Sequence has been completed. ② Confirm that KCV-3422 in the 50% NaOH line from OSBL, KCV-3423 in the 10S line and KCV-3424 in the DIW line have been closed. ③ Compare the indication on LIK-3421 with that of LG-3421. ④ Close all block valves mentioned in the item ② above.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
35. Adsorber (TT-1131 A/B)	1) <u>Differential pressure</u> Upper Limit (PIC-3131) (PIK-3132A/B)	0.1 > kg/cm ² G	(1) 0.5 kg/cm ² G	(2) 1.0 kg/cm ² G	As the silica gel and active carbon, which are filled in the adsorber, are pulverized to fine particles, the differential pressure across the adsorber column increases. Accordingly, it is specified to prevent a fine particles from mingling with PTA product due to their entrainment with the waste gas.	Pressure Indicator with Alarm (PIC-3131) (PIK-3132A/B)	(1)-① Check the flow rate of the purified waste gas and adjust it, if necessary. ② Check the silica gel and active carbon for how long they have been used. ③ Check to see if there is not any symptom of breakage of the screen supporting silica gel and active carbon. (2)-① Stop the operation of adsorber, and replace SG and AC with new ones.
	2) <u>Regeneration Temperature</u> (TIK-3131)	60°C ≤	60°C	60°C >	A lower temperature of heat regeneration gas reduces the life of the silica gel and causes an insufficient regeneration.	Temperature Indicator with Alarm	① Confirm TIK-3131 is functioning properly. ② Check the flow rate of the purified gas to be used for regeneration of SG and AC, and adjust it, if necessary. ③ Check the temperature on TG-3124 on the outlet of the Regeneration Gas Heater TE-1131. If it is lower than the specified temperature, check the flow rate of steam fed to the heater and also, check the working condition of the steam trap. ④ Check the flow rate of the purified waste gas. (Note) If the regeneration temperature doesn't rise up to 60°C within 8 hours, the operation shall be put into "Forced Cooling" by the Sequence.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
36. WG Filter (TM-1121 A/B)	<u>Differential Pressure</u> Upper Limit (PDI-3133)	0.06 kg/cm ² G	0.4 kg/cm ² G (Alarm)	0.5 kg/cm ² G	It is specified to prevent the filter element from being collapsed due to its plugging.	DP Indicator with Alarm (PDI-3133)	① Switch the filter over to the stand-by side and clean it.
37.Oil Separator (TU-800)	1) <u>Liquid Level</u> (1)Upper Limit (LIC-1800)	25~75%	89%	95%	(1)A high level of the oil separator leads to a deterioration of separating oil from water.	Level Indicator and Regulator with Alarm (LIC-1800)	(1) ① Adjust the liquid level by controlling FICQ-1800 on manual. ② Confirm FICV-1800 is functioning properly. ③ Compare the indication on LIC-1800 with the actual level in the field. ④ Check to be sure that TP-800 is pumping properly. ⑤ Check the flow rate on FICQ-1800, and start spare pump if the level-up of O.S. is due to an increase in the waste water. ⑥ In case where the level of O.S. can not be returned to the normal level, switch the waste water over to the Blow-Down Pit.
	(2)Lower Limit (LIC-1800)	25~75%	20% (Alarm)	15%	It is specified to prevent the pump from getting into the cavitation.		(2)-① Adjust an amount of waste water to be sent to the waste water treatment unit by controlling FICQ-1800 on manual. ② Confirm that FICV-1800 is functioning properly. ③ Compare the indication on LIC-1800 with the actual level in the field.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TU-800)	2) <u>PH</u> (1) Upper Limit (AI-1800)	7~9	10.5 (Alarm)	11	It is specified to prevent an abnormal waste water from being sent to the waste water treatment unit.	PH Indicator and Regulator with Alarm (AIC-1800)	(1), (2) ① Check the working condition for the PH preparation system, and adjust FIC-3423 and FIC-3425, if the PH so prepared is out of the specified range. (Note) FIC-3423: 50% NaOH FIC-3425: 3% NaOH ② Check the PH on AI-1803 installed at the inlet of the Oil Separator. ③ Check the PH of the waste water for TA and PTA units separately, and switch the abnormal waste water over to the Blow-Down Pit. a) PH of waste water for TA unit; AI-1804 b) PH of waste water for PTA unit; AI-1806 ④ If the PH of the O.S. could not be adjusted to the allowable value, switch the destination of the waste water from the waste water treatment unit to the dirty waste water on the outlet of TP-800.
	(2) Lower Limit (AI-1800)	7~9	6 (Alarm)	5.5			
	3) <u>Temperature</u> Upper Limit (TI-1800)	50~70°C	73°C (Alarm)	75°C	Temperature of the waste water has an influence on the activity of the active sludge.	Temperature Indicator with Alarm (TI-1800)	① Check to see if there is not any waste water with high temperature getting into O.S. ② Check the temperature of waste water to be discharged from PW-901, and reset the revolution of fan (PC-901), if necessary. ③ Inform A.S. plant of the abnormality of the waste water.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
38.Steam Flash Drum (TD-1201/ 1203/ 1210)	1) <u>Liquid Level</u> Upper Limit (LICS-3201)	50%	(1) 65% (Alarm)	(2) 80% Interlock	In the steam flash drum, an excess amount of 10SC, and 3SC are sent to TD-1203 from TD-1210, and to TD-1201 from TD-1203 respectively by the pressure difference. However, if each SC pump becomes impossible to transport SC, TD-1201 is full of SC because 10SC and 3SC are sent to TD-1201. And so, when the level of TD-1201 reaches the specified level, SC in TD-1201 is blown to the sewer through SDV-3210 by the actuation of the Interlock.	Level Indicator and Regulator with Alarm and Interlock (LICS-3201)	(1)-① Compare the indication on LICS-3201 with that of LG-3211. ② Check an amount of LPW make-up on FI-3227 and the working condition of LCV-3201A. ③ Check an amount of SC sent to the OSBL (FIQ-3226) and the working condition of LCV-3201B. ④ Confirm that the SC pump TP-1201 is pumping up properly. (2)-① Upon reaching 80% on LICS-3201, SC is blown through SDV-3210 to the sewer. (Note) When the level on LICS-3201 returns to the normal level, the said Interlock is reset and SDV-3210 is closed.
39.PA Compressor Train (TC-201) (TB-1131) (TB-1132) Motor/ Generator	1) <u>LO pressure</u> Lower Limit (PI-1262) (PS-1263) (PI-1261)	4.5kg/cm ² G	3.5kg/cm ² G (Alarm) a) 1.2 kg/cm ² G Interlock	2.5kg/cm ² G b) 2.0 kg/cm ² G Interlock	Lube Oil Unit lubricates forcibly and protects the compressor, speed gear, steam turbine and gas turbine. They are large, high-speed equipment without the spare units so that their damages give a critical impact on the operation.	Pressure Indicator with Alarm (PI-1262)	1)-a) When LO pressure on PS-1263 falls down to 1.2kg/cm ² G, spare LO pump is started automatically by the Interlock . 1)-b) When LO pressure on PS-1261 falls down to 0.5kg/cm ² G, steam turbine and GET stop, compressor gets into the Blow-off condition.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TC-201) (TB-1131) (TB-1132)	2) <u>LO Temperature</u>						
	(1) Upper Limit (TCV-1257)	45°C	55°C (Alarm)	60°C	A temperature rise of LO lowers the cooling efficiency on the bearing. On the contrary, a temperature decrease of LO increases the viscosity of LO, resulting in giving a bad influence on the machine.	Temperature Indicator and Regulator with Alarm (TIS-1257) (TIA-1258)	(1), (2) ① Check the flow rate of cooling water fed to TE-257 (FI-3559) and check it by controlling TV-1257 to control the temperature of LO. ② Confirm that TCV-1257 is functioning properly. ③ Adjust by the flow rate of cooling CCW fed TE-257 and the opening TV-1257.
	(2) Lower Limit (TCV-1257)	45°C	30°C (Alarm)				
	(TIA-1258)		30°C (Starting Condition)	30°C			
	3) <u>Level of LO</u>						
	Lower Limit (LA-1255)	8.3m ³	7.44m ³ (Alarm)	2.8m ³	A fall of oil level in the LO tank, which means lowering of residence time of LO, becomes impossible of suppressing a generation of foam and results in a deterioration of lubrication performance. Also, the mentioned above leads to a damage of gear pump due probably to a cavitation.	Fixed Level Instrument with Alarm (LA-1255)	① Confirm the oil level by the LG installed on the LO tank. ② Replenish the LO tank with the new oil. ③ Check to be sure that there is not any oil leaked around the LO unit. If a lot of oil is leaking around the LO unit, move to the ESD.
	4) <u>Bearing Temperature</u>						
	Upper Limit	°C	°C (Alarm)	°C	They are specified to prevent each bearing from seizing due to high temperature.	Temperature Indicator with Alarm (TV-1270 ~ TIS-1286) *Note: TI-1272A/B	① Confirm that the LO supply temperature indicated on TCV-1257 is within the specified one, and that TCV-1257 is functioning properly. ② Check the Lo temperature of LO tank.
	(1) Steam Turbine (TI-1270 ~ 1272A/B)	75~90	(0) 100 (1) 100 (2) 100 (3) 100	115			

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TC-201) (TB-1131) (TB-1132)	(2)Speed Gear (TI-1273~1274 TI-1277~1278) (3)Compressor (HPC) (TJI-1279~1280) (4)Compressor (LPC) (TJI-1275~ 1276) (5)GET (TI-1283~1286) (6)Motor (TI-1281~1280)	°C 70~95 70~95 70~95 70~90 80	°C (Alarm) 105 105 105 105 85	°C 115 115 115 110 85			③ Confirm that the LO pump TP-251A(B) is functioning properly. ④ Confirm that there is no clogging in the oil filter. ⑤ Confirm that the LO pressure on each bearing is within the specified figure. ⑥ Check an amount of oil returning from each bearing to the LO tank through SG. ⑦ Inform the Maintenance Div. of the abnormal temperature rise for the bearing. ⑧ In case of a steep temperature rise for the bearing, move to the ESD.
	5) <u>Shaft Displacement</u> Upper Limit	mm	a)mm (Alarm)	b)mm Interlock	They are specified to prevent the compressor train from being damaged due to the shaft displacement.	Shaft Displacement Indicator with Alarm and Interlock (ZIS-1260 ~ ZIS-1263)	a)-① Check the temperature and vibration on each bearing. ② Inform the Maintenance Div. of the abnormal shaft displacement and receive their instruction on the countermeasures to be taken against it. b)-① According to the actuation of the Interlock, the steam turbine and GET stop, and compressor gets into the blow-off condition.
	(1)Steam Turbine (ZIS-1260) (2)Compressor (Main Shaft) (ZIS-1261) (3)Compressor (Turbine Shaft) (ZIS-1262) (4)GET (ZIS-1263)	+0.2~0.4 +0.1 ~ -0.4 +0.2 ~ -0.6 +0.1 ~ -0.4 ± 0.20	+0.2 -0.4 +0.1 -0.4 +0.1 -0.4	+0.3 -0.4 +0.20 -0.45 +0.2 -0.75 +0.2 -0.5			

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TC-201) (TB-1131) (TB-1132)	6) <u>Shaft Vibration</u> Upper Limit (1) Steam Turbine (VI-1260A/B) (2) Speed Gear a) Turbine Shaft (VI-1261A/B) b) Shaft (VI-1262A/B) (3) Compressor (HPC) (VI-1263A/B) (4) Compressor (LPC) (VI-1264A/B) (5) GET (VI-1265A/B)	μ m 6> 9> 21> 7> 7> 4>	a) μ m (Alarm) 95 95 76 45 56 45	b) μ m 140 140 111 67 82 66	They are specified to prevent the compressor train from being damaged due to the high shaft vibration.	Shaft Vibration Indicator with Alarm (VI-1260A/B ~ VI-1265A/B)	a)-① Check the actual vibration on the specified point by the handy-vibrometer. ② Inform the Maintenance Div. of the abnormal shaft vibration and receive their instruction on the countermeasures to be taken against it. ③ Prepare for both the load-down and the shutdown of the plant. b)-① When the shaft vibration reaches the critical figure, immediately stop the compressor train.
	7) Motor/Generator (TIS-1290~1293)	120°C	125°C	130°C	They are specified to prevent the Motor and Generator coil from damaged due to the high temperature.	Motor coil indicator with Alarm (TIS-1290~1292)	c)-① Check the air filter of the motor whether clog or not. ② Inform the maintenance Div. Of the abnormal coil of the motor and receive their instruction on the counter-measure to be taken against it. ③ Take the same actions as mentioned in the item 6)-a)-③. ④ When the temperature reaches the critical figure, immediately stop the compressor train.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
40.PA Compressor (TC-201)	1) <u>DP on Air Filter</u> Upper Limit (PDI-1255)	30~55 mmAq	63mmAq (Alarm)	120mmAq	It is specified to prevent the air filter from being damaged due to a rise of the differential pressure for the air filter.	DP Indicator with Alarm (PDI-1255)	① Compare the indication on PDI-1255 with those on PDI-1255A/B in the field. ② Check the working condition for the automatic roll-up device of filter element. ③ Roll up the filter element a little depending on the clogging condition for the filter which can be observed through the sight glass. ④ In case where the filter element has been rolled up to the end, replace it with the new one.
	2) <u>Pressure on Last Stage of HPC</u> Upper Limit (PIC-1260)	12.0 ~ 12.6 kg/cm ² G	a) 13.0 kg/cm ² G (Alarm)	b) 15.0 kg/cm ² G (Design)	It is specified to prevent the reactor system designed by 15 kg/cm ² G from being damaged due to overpressure.	Pressure Indicator and Regulator with Alarm (PIC-1260)	a)-① In case of constant pressure control, check the set value on PIC-1260 and the degree of angle on the static blade for LPC indicated on PIC-1260A & B and HPC indicated on PIC-1260C & D, and confirm that the pressure is controlled at the specified figure. ② In case of constant flow control, check the set value on FIC-1201 and the degree of angle on the static blade for LPC indicated on PIC-1260A & B and HPC indicated on PIC-1260C & D, and confirm that the flow is controlled at the specified figure. ③ Check the reactor pressure on PICK-1202 and the process flow rate on FIC-1201. ④ Confirm that the blow-off valve PV-1250 is functioning properly.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TC-201)							b)-① In case where the pressure on PIC-1260 exceeded the upper limit or the compressor got into the serging condition, immediately stop the compressor train by actuating the shutdown switch.
41.GET (TB-1131)	1)Gas Temperature on outlet of 1st Stage GET				(1)Gas temperature on the outlet of 1st stage GET is specified to be below the designed temperature for TE-1133.	Temperature Indicator with Alarm (TI-3133)	(1), (2) ① Check the flow rate of WG fed to GET on FIC-3102 and the pressure on PI-3102. ② Check the temperature of WG fed to the 1st stage GET indicated on TI-3132, and check the flow rate of 3 ^k steam to TE-1132.
	(1)Upper Limit (TI-3133)	75~80°C	100°C (Alarm)	140°C			③ Check to be sure that the steam trap of TE-1132 is functioning properly.
	(2)Lower Limit (TI-3133)	75~80°C	32°C (Alarm)	30°C	(2)Turbine blade is corroded with a liquid liquefied below the dew point of the WG fed to the GET.	Ditto	④ Check the pressure of WG on the outlet of 1st stage GET. (PI-3106)
	2)Gas Temperature on Outlet of 2nd Stage GET				(1)It is specified to prevent the equipment from being damaged due to a temperature rise.	Temperature Indicator with Alarm (TI-3135)	(1), (2) ① Check the flow rate of WG fed to GET on FIC-3102. ② Check the pressure of WG fed to 2nd stage GET. (PI-3106 and PI-3108)
	(1)Upper Limit (TI-3135)	40~45°C	100°C (Alarm)	140°C			③ Check the temperature of WG fed to 2nd stage GET on TI-3134, and check the flow rate of 15K steam to TE-1133.
	Seal-Air Flow (FI-1250)	72Nm ³ /hr		140Nm ³ /hr			④ Check to be sure that the steam trap of TE-1133 is functioning properly.
	(2)Lower Limit (TI-3135)	40~45°C	12°C (Alarm)	10°C	(2)It is specified due to the reason as mentioned in the item 1)-2) above.	Temperature Indicator with Alarm (TI-3135)	

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TB-1131)	3) <u>Over Speed</u> Upper Limit (SIS-1261)	1000rpm	1100rpm (Alarm)	1252rpm Interlock	They are specified to prevent the compressor train from being damaged due to the overspeed of Expander.	Revolution indicator with Alarm and Interlock	a)Decrease the revolution speed to 1100rpm or lower, because it is the allowance maximum speed. b)At the critical speed, the Expander turbine is stopped by the actuation of the Interlock.
42.Steam Turbine (TB-1132)	1) <u>Overspeed</u> Upper Limit (SIS-1260)	4,160 ~ 4,170 rpm	a) 4,382rpm (Alarm)	b) 4,957rpm Interlock	They are specified to prevent the compressor train from being damaged due to the overspeed of steam turbine.	Revolution Indicator with Alarm and Interlock (SIS-1260)	a)Decrease the revolution speed to 4,382rpm or lower, because it is the allowable maximum speed. b)At the critical speed, the steam turbine is stopped by the actuation of the Interlock.
	2) <u>Turbine Exhaust Pressure</u> Upper Limit (PIS-3140) (PS-3254)	-662mmHg	a) -588mmHg (Alarm) -600mmHg (Starting Condition)	515mmHg Interlock	A rise of exhaust pressure leads to the power shortage generated by the steam turbine, resulting in an inoperability of the compressor train and the damage of equipment.	Pressure Indicator with Alarm (PIS-3140) Pressure Switch with Interlock (PIS-3140)	a)-① Check the pressure of steam being used for 1st and 2nd steam ejectors. ② Switch the steam ejector over to the stand-by side. ③ Confirm that the steam trap of ejector condenser (TE-1135) is functioning properly. ④ Check to be sure that there is not any equipment in the vacuum system to suck in air from the atmosphere. ⑤ Check to be sure that the flow rate of CCW fed to UE-134 on FIQ-3380 is the specified figure.
	3) <u>Turbine Exhaust Temperature</u> ()	49°C	150°C	200°C	Ditto	Temperature Indicator with Alarm and Interlock ()	

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TB-1132)							⑥ In case where the flow rate of PA to the reactor can not be secured, decrease the flow rate of PX fed to the reactor or prepare for the unit shutdown taking into consideration the situation. b) At the exhaust pressure of -400mmHg, steam turbine is automatically stopped by the actuation of the Interlock.
	4) <u>Liquid Level in TE-1134</u> (1) Upper Limit (LIC-3135) (LIS-3134)	50%	a) 60% (Alarm)	b) 71.4% Interlock	A rise of liquid level in TE-1134 becomes an insufficient cooling of steam exhausted from the steam turbine and a rise of exhaust pressure, thereby resulting in inoperability of the steam turbine.	Level Indicator and Regulator with Alarm (LICS-3134, LIC-3135) Level Switch with Interlock (LIS-3134)	1)-a)-① Compare the indications on LIS-3134 and LIC-3135 with that of LG-3134. ② Confirm that FV-3134 and LV-3135 are functioning properly. ③ Confirm that TP-1131 is pumping properly. ④ Check to be sure that there is not any leakage of CCW due probably to the breakage of tube in TE-1134. (1)-b)-① Stand-by TP-1131 is automatically started by the actuation of Interlock.
	(2) Lower Limit (LIC-3135) (LIS-3134)	50%	40% (Alarm)	28.6% (Alarm)	A fall of liquid level in TE-1134 may lead to a damage of equipment and an insufficient cooling of TE-1135 by causing TP-1131 to generate the cavitation.	Level Indicator and Regulator with Alarm (LIS-3134, LIC-3135)	(2)-① Compare the indications on LIS-3134 and LIC-3135 with that of LG-3134. ② Confirm that FV-3134 and LV-3135 are functioning properly. ③ Check the flow rates of 1BW to TE-201B and TE-201C. (FI-1222/FI-1223) ④ Replenish TE-1134 with fresh DIW.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
43. Catalytic Oxidizer (TD-901)	1) <u>Temperature</u> (1) Upper Limit (TIC-1904)	375°C	483°C (Alarm)	550°C (Design)	The upper limit on the temperature is set at the design temperature of TD-901.	Temperature Controller with Alarm (TIC-1904)	1)-① Check the flow rate and the temperature of WG fed to TD-901. ② Check the combustible gas level on AI-1208. ③ IF temperature control cannot be achieved by adjusting fuel gas to fired heater TF-901, the system should be by-passed and purged.
	(2) Lower Limit (TIC-1904)	375°C	350°C (Alarm)	316°C Interlock	A lower temperature of catalytic oxidizer precipitates Bromine in the bed, damaging catalyst.	Temperature controller with Alarm (TIC-1904)	2)-① Check the functioning of TE-901 & TE-902. ② Check the flow rate and the temperature of WG FED to TD-901. ③ IF temperature control cannot be achieved by adjusting fuel to fired heater TF-901, the system should be by-passed and purged.
44. Fired Heater (TF-901)	1) <u>Temperature</u> (1) Upper Limit (TIC-1904)	375°C	494°C (Alarm)	550°C (Design)	(1) It is specified to prevent the heating tube in TF-901 from being overheated and the catalyst in TD-901 from being damaged due to abnormal temperature rise of WG.	Temperature Indicator and Regulator with Alarm and Interlock (TIC-1904)	(1)-① Adjust the flow rate of FG fed to PF-901. ② Check the flow rate of WG fed to PF-901. ③ Check the combustible gas level of AI-1208. ④ If the temperature control cannot be achieved by adjusting fuel gas to fired heater TF-901, the system should be by-passed and purged.
	(2) Lower Limit (TIC-1904)	375°C	350°C (Alarm)	338°C	(2) It is specified to prevent the Ca/Mn catalyst in TD-901 from being precipitated Bromine in the bed and damaged.	Ditto	(2) Ditto

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TF-901)	2) <u>Pressure of FG</u> (1) Lower Limit (PIS-1909)	0.2kg/cm ² G	kg/cm ² G (Alarm)	kg/cm ² G Interlock	It is specified to prevent the main burner from extinguishing as well as to avoid gas expression followed by the extinction of burner due to a fall of the FG pressure.	(1) FG Pressure Indicator with Alarm and Interlock (PIS-1909)	a)-① Check the indicator on FICQ-3911. (Flow rate of FG) ② Check the indication on PC-3911 and working condition of PV-3911. ③ Confirm that pilot burners are ignited properly. b)-① FG main burners are extinguished by the actuation of the Interlock. ② Immediately switch the operation over to the by-pass and purging the system.
45. Vent Scrubber (TT-901)	1) <u>Bottom Section Liquid Level</u> (1) Upper Limit (LIC-1901A)	30%	50% (Alarm)	80%	A rise of liquid level in TT-901 worsen the gas-liquid separation efficiency, thereby increasing a mist entrained with WG. There is a possibility that liquid in Tt-901 is spouted up to the tray section and damaged the tray.	Level Indicator and Regulator Alarm (LIC-1901A)	(1)-① Adjust the liquid level by controlling LIC-1901A on manual. ② Check LCV-1901A for its function. ③ Check an amount of liquid discharged through LV-1901A 6 FV-1904.
	(2) Lower Limit (LIC-1901A)	30%	105 (Alarm)	10%	(2)-1. It is specified to prevent the pump from getting into the cavitation condition. 2. The said above leads to an insufficient quenching of WG from TE-901.		(2)-① Check the indicator on LIC-1901A for its correctness by comparing it with that on LG-1901A.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(TT-901)	2) <u>Top Section Liquid Level</u> (1) Upper Limit (LIC-1901B)	30%	505 (Alarm)	80%	A rise of liquid level in TT-901 worsen the gas-liquid separation efficiency, thereby increasing a mist entrained with WG. (2)-1. It is Specified to prevent the pump from getting into the cavitation condition. 2. The said above leads to an insufficient scrubbing of WG.	Level Indicator and Regulator with Alarm (LIC-1901B)	① Adjust the liquid level by controlling LIC-1901B on manual. ② Check LCV-1901B for its function. ③ Check an amount of liquid discharged through LV-1901B & LV-1904. ④ Check the temperature of TT-901 on TI-1908.
	3) <u>PH</u> (1) Lower Limit (AI-1906)	9	7~8	7.0	Decrease of PH in TT-901 worsen the efficiency of the bromine removal and disturb to minimize corrosion from dissolved bromine in the liquid stream.	PH Indicator and Regulator with Alarm (AI-1906)	① Adjust the PH by controlling FIC-3424 on manual. ② Adjust the flow rate of sodium formate.
46. <u>Sodium Formate Drum</u> (TD-1900)	<u>Liquid Level</u> Upper Limit (LIK-1900)	10~90%	95% (Alarm)	95%	It is specified to prevent Sodium Formate causing dermatitis from overflowing due to an excessive rise of liquid level.	Level Indicator with Alarm (LIK-1900)	① Confirm that Sodium Formate throwing by the person has been completed. ② Confirm that Sodium Formate preparation sequence has been completed. ③ Confirm that KCV-3902 installed on the DIW receiving line has been closed. ④ Check the indication on LIK-1900.

Equipment	Factor	Operation Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
47. <u>Sodium</u> <u>Formate</u> <u>Measuring</u> <u>Drum</u> (TD-1901)	<u>Liquid Level</u> Upper Limit (LIK-3901)	10~90%	95% (Alarm)	95%	It is specified to prevent Sodium Formate causing dermatitis from overflowing due to an excessive rise of liquid level.	Level Indicator with Alarm (LIK-1901)	① Confirm that Sodium Formate Receiving in the Sequence has been completed from TD-1900 to TD-1901. ② Confirm that KCV-3901 installed on the Sodium Formate receiving line has been closed. ③ Compare the indication on LIK-3901 with that of LG-3901.